Bell-State Measurement Simulator

A Prelude to Entanglement Swapping and Quantum Networking

Artifact class: Educational brief, theory companion, guided activity, and assessment module for the Bell-state measurement simulator.

Purpose: Introduce Bell states as entangled resources, explain the Bell-state measurement construction, connect BSM to quantum networking, and prepare learners for the state-vector-based entanglement swapping simulator.

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Abstract

Entanglement is one of the defining resources of quantum information science. In an entangled state, the joint quantum state of two systems contains information that cannot be reduced to independent states of the individual parts. Bell states are the simplest and most important examples: they are maximally entangled two-qubit states and serve as basic resources for teleportation, entanglement swapping, and quantum networking. This brief introduces Bell-state measurement (BSM) through a compact simulator. The simulator allows a user to select one of four Bell states, apply the inverse Bell-preparation circuit, and measure both qubits in the computational basis. The resulting two-bit outcome identifies the selected Bell state with 100% certainty. This activity prepares students for entanglement swapping by showing how Bell-basis information is converted into classical measurement information.

> **Core idea.** Bell states are entangled two-qubit states. Their information is stored in the joint state of the pair, not in either qubit alone. A Bell-state measurement converts this entangled Bell-basis information into ordinary two-bit classical information. For an isolated Bell state, this identifies which Bell state was given. In entanglement swapping, the same measurement logic is applied to the middle pair of a four-qubit state, and the two-bit outcome labels which entangled state has been created between two remote outer qubits.

1. Introduction: Entanglement, Bell States, and Quantum Networks

Entanglement is the feature of quantum mechanics that makes quantum information fundamentally different from ordinary classical information. When two systems are entangled, the state of the pair cannot be understood as two separate states carried independently by each subsystem. Instead, the physically meaningful object is the joint state.

The simplest examples are the Bell states. For example,

$$|\Phi^+\rangle = \frac{1}{\sqrt{2}}\left(|00\rangle + |11\rangle\right).$$

This state says that if both qubits are measured in the computational basis, the results are perfectly correlated: the outcome is either 00 or 11. However, it is not merely a classical statement that the system is secretly either $|00\rangle$ or $|11\rangle$. It is a coherent quantum superposition of both possibilities. The relative phases between the terms matter, and those phases distinguish different Bell states.

This is why Bell states are central to quantum communication. They provide shared nonclassical correlations between separated systems. Those correlations can be used for teleportation, entanglement swapping, quantum repeaters, distributed sensing, and networked quantum computation.

A central operation in quantum networking is **entanglement swapping**. Suppose node A is entangled with node B, and node C is entangled with node D. If the middle qubits B and C are jointly measured in the Bell basis, then the outer qubits A and D can become entangled, even if they never directly interacted.

The key operation is the **Bell-state measurement**, abbreviated BSM. The present simulator focuses only on the BSM itself. The user chooses one of four Bell states, the simulator applies the BSM transformation, and then both qubits are measured in the computational basis. The result identifies the original Bell state with 100% certainty.

This is a useful prelude to entanglement swapping because it isolates the measurement logic before embedding that logic into a larger four-qubit network process.

2. Learning Objectives

After completing this activity, students should be able to:

1. Write the four Bell states in computational-basis notation.
2. Explain what it means for a two-qubit state to be entangled.
3. Explain why Bell states are maximally entangled two-qubit states.
4. Explain why the information in a Bell state is stored in joint correlations.
5. Represent a CNOT gate using tensor-product operator notation.
6. Explain why a Bell-state measurement can be viewed as running Bell-state preparation backward.
7. Compute how the inverse Bell-preparation circuit maps Bell states to computational-basis states.
8. Interpret the two-bit BSM outcome as a Bell-state label.
9. Explain why this measurement logic is needed for entanglement swapping and quantum networking.

3. The Computational Basis

For one qubit, the computational basis states are

$$|0\rangle = \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \qquad |1\rangle = \begin{pmatrix} 0 \\ 1 \end{pmatrix}.$$

For two qubits, the computational basis is

$$|00\rangle\,, \quad |01\rangle\,, \quad |10\rangle\,, \quad |11\rangle\,.$$

In tensor-product notation,

$$|00\rangle = |0\rangle \otimes |0\rangle\,, \qquad |01\rangle = |0\rangle \otimes |1\rangle\,,$$

$$|10\rangle = |1\rangle \otimes |0\rangle\,, \qquad |11\rangle = |1\rangle \otimes |1\rangle\,.$$

The ordering used in the simulator is

$$|00\rangle\,,\ |01\rangle\,,\ |10\rangle\,,\ |11\rangle\,.$$

4. Bell States and Entanglement

The four Bell states are

$$|\Phi^+\rangle = \frac{1}{\sqrt{2}}\left(|00\rangle + |11\rangle\right),$$

$$|\Phi^-\rangle = \frac{1}{\sqrt{2}}\left(|00\rangle - |11\rangle\right),$$

$$|\Psi^+\rangle = \frac{1}{\sqrt{2}}\left(|01\rangle + |10\rangle\right),$$

$$|\Psi^-\rangle = \frac{1}{\sqrt{2}}\left(|01\rangle - |10\rangle\right).$$

These states form an orthonormal basis for the two-qubit Hilbert space. They are also maximally entangled states.

To see what this means, consider

$$|\Phi^+\rangle = \frac{1}{\sqrt{2}}\left(|00\rangle + |11\rangle\right).$$

If the first qubit is measured in the computational basis, the result is 0 with probability 1/2 and 1 with probability 1/2. The same is true for the second qubit. Each qubit alone looks random.

But the pair is not random in the same way. The measurement outcomes are perfectly correlated:

$$q_1 = 0 \Rightarrow q_2 = 0, \qquad q_1 = 1 \Rightarrow q_2 = 1.$$

The state therefore contains information in the correlations between the two qubits, not in either qubit separately.

This is the signature of entanglement. The joint state cannot be factored into a product of single-qubit states:

$$|\Phi^+\rangle \neq |\psi\rangle_1 \otimes |\phi\rangle_2 .$$

A quick way to see this is to try writing

$$|\psi\rangle_1 = a\,|0\rangle + b\,|1\rangle\,, \qquad |\phi\rangle_2 = c\,|0\rangle + d\,|1\rangle\,.$$

Then

$$|\psi\rangle_1 \otimes |\phi\rangle_2 = ac\,|00\rangle + ad\,|01\rangle + bc\,|10\rangle + bd\,|11\rangle\,.$$

To equal $|\Phi^+\rangle$, we would need

$$ac = \frac{1}{\sqrt{2}}, \qquad bd = \frac{1}{\sqrt{2}},$$

but also

$$ad = 0, \qquad bc = 0.$$

These conditions cannot all be satisfied at the same time. Therefore $|\Phi^+\rangle$ is not a product state.

This matters because entanglement is a resource. It enables quantum protocols that have no purely classical equivalent. In quantum networking, Bell states can connect remote systems, support teleportation, and enable entanglement swapping across intermediate nodes.

However, using Bell states requires knowing *which* Bell state is present. That is the role of a Bell-state measurement.

Why ordinary measurement is not enough. If we measure $|\Phi^+\rangle$ and $|\Phi^-\rangle$ directly in the computational basis, both give the same possible outcomes:

$$00 \quad \text{or} \quad 11.$$

The difference between them is the relative sign:

$$|\Phi^+\rangle = \frac{1}{\sqrt{2}}(|00\rangle + |11\rangle), \qquad |\Phi^-\rangle = \frac{1}{\sqrt{2}}(|00\rangle - |11\rangle).$$

That sign is a quantum phase relationship, not something revealed by simply measuring each qubit separately in the computational basis. A Bell-state measurement first changes basis so that the Bell-state label becomes an ordinary two-bit outcome.

5. Bell-State Preparation

One standard way to prepare Bell states is to start with computational-basis states, apply a Hadamard gate to the first qubit, and then apply a CNOT gate with the first qubit as control and the second qubit as target.

The Hadamard gate is

$$H = \frac{1}{\sqrt{2}}\begin{pmatrix}1 & 1\\ 1 & -1\end{pmatrix}.$$

It acts as

$$H\left|0\right\rangle = \frac{1}{\sqrt{2}}\left(\left|0\right\rangle + \left|1\right\rangle\right) = \left|+\right\rangle, \qquad H\left|1\right\rangle = \frac{1}{\sqrt{2}}\left(\left|0\right\rangle - \left|1\right\rangle\right) = \left|-\right\rangle.$$

The Pauli X gate is

$$X = \begin{pmatrix}0 & 1\\ 1 & 0\end{pmatrix},$$

with

$$X\left|0\right\rangle = \left|1\right\rangle, \qquad X\left|1\right\rangle = \left|0\right\rangle.$$

The CNOT gate, with qubit 1 as control and qubit 2 as target, can be written in tensor-product operator notation as

$$\mathrm{CNOT} = \left|0\right\rangle\left\langle 0\right| \otimes I + \left|1\right\rangle\left\langle 1\right| \otimes X.$$

This means

$$\mathrm{CNOT}\left|00\right\rangle = \left|00\right\rangle, \qquad \mathrm{CNOT}\left|01\right\rangle = \left|01\right\rangle,$$
$$\mathrm{CNOT}\left|10\right\rangle = \left|11\right\rangle, \qquad \mathrm{CNOT}\left|11\right\rangle = \left|10\right\rangle.$$

The Bell-preparation unitary is

$$U_{\mathrm{prep}} = \mathrm{CNOT}(H \otimes I).$$

For example,

$$(H \otimes I)\left|00\right\rangle = H\left|0\right\rangle \otimes I\left|0\right\rangle = \frac{1}{\sqrt{2}}(\left|0\right\rangle + \left|1\right\rangle) \otimes \left|0\right\rangle,$$

so

$$(H \otimes I)\left|00\right\rangle = \frac{1}{\sqrt{2}}(\left|00\right\rangle + \left|10\right\rangle).$$

Now apply CNOT:

$$\mathrm{CNOT}\frac{1}{\sqrt{2}}(\left|00\right\rangle + \left|10\right\rangle) = \frac{1}{\sqrt{2}}(\left|00\right\rangle + \left|11\right\rangle).$$

Thus,

$$U_{\mathrm{prep}}\left|00\right\rangle = \left|\Phi^+\right\rangle.$$

6. Bell-State Measurement as the Reverse Process

A Bell-state measurement asks:

Which Bell state is this two-qubit system in?

Mathematically, we can answer this question by applying the inverse Bell-preparation circuit and then measuring in the computational basis.

Because

$$H^\dagger = H, \qquad \mathrm{CNOT}^\dagger = \mathrm{CNOT},$$

the inverse of

$$U_{\text{prep}} = \text{CNOT}(H \otimes I)$$

is

$$U_{\text{BSM}} = U_{\text{prep}}^{\dagger} = (H \otimes I)\text{CNOT}.$$

So the BSM simulator applies

$$\boxed{U_{\text{BSM}} = (H \otimes I)\text{CNOT}}$$

and then measures both qubits in the computational basis.

7. Worked Example: Applying BSM to $|\Phi^+\rangle$

Start with

$$|\Phi^+\rangle = \frac{1}{\sqrt{2}}(|00\rangle + |11\rangle).$$

Apply CNOT:

$$\text{CNOT}\,|\Phi^+\rangle = \frac{1}{\sqrt{2}}\left(\text{CNOT}\,|00\rangle + \text{CNOT}\,|11\rangle\right).$$

Using

$$\text{CNOT}\,|00\rangle = |00\rangle\,, \qquad \text{CNOT}\,|11\rangle = |10\rangle\,,$$

we get

$$\text{CNOT}\,|\Phi^+\rangle = \frac{1}{\sqrt{2}}(|00\rangle + |10\rangle).$$

Factor this as

$$\frac{1}{\sqrt{2}}(|00\rangle + |10\rangle) = \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle) \otimes |0\rangle\,.$$

Therefore,

$$\text{CNOT}\,|\Phi^+\rangle = |+\rangle \otimes |0\rangle\,.$$

Now apply $H \otimes I$:

$$(H \otimes I)(|+\rangle \otimes |0\rangle) = H\,|+\rangle \otimes I\,|0\rangle\,.$$

Since

$$H\,|+\rangle = |0\rangle\,,$$

we obtain

$$(H \otimes I)\text{CNOT}\,|\Phi^+\rangle = |0\rangle \otimes |0\rangle = |00\rangle\,.$$

Thus,

$$\boxed{|\Phi^+\rangle \longrightarrow |00\rangle \longrightarrow \text{measurement outcome } 00.}$$

8. Full Bell-State Measurement Mapping

Applying the same logic to all four Bell states gives

$$\boxed{\begin{aligned} |\Phi^+\rangle &\longrightarrow |00\rangle\,, \\ |\Phi^-\rangle &\longrightarrow |10\rangle\,, \\ |\Psi^+\rangle &\longrightarrow |01\rangle\,, \\ |\Psi^-\rangle &\longrightarrow |11\rangle\,. \end{aligned}}$$

Then measuring both qubits gives a two-bit outcome.

Input Bell state	After inverse Bell circuit	Measured bitstring
$\lvert\Phi^+\rangle$	$\lvert 00\rangle$	00
$\lvert\Phi^-\rangle$	$\lvert 10\rangle$	10
$\lvert\Psi^+\rangle$	$\lvert 01\rangle$	01
$\lvert\Psi^-\rangle$	$\lvert 11\rangle$	11

The measurement outcome identifies the input Bell state with 100% certainty because the four Bell states are orthogonal and the inverse Bell circuit maps them to distinct computational-basis states.

> **Important interpretation.** For an isolated input Bell state, the BSM tells us which Bell state was present. For entanglement swapping, the middle pair is not merely an isolated pre-existing Bell state. Instead, the BSM projects the middle pair onto one Bell-state outcome, and that classical outcome labels which Bell state the outer qubits have become.

9. Connection to Entanglement Swapping

Now consider four qubits. Suppose qubits 1 and 2 are initially entangled, and qubits 3 and 4 are initially entangled:

$$|\Phi^+\rangle_{12} \otimes |\Phi^+\rangle_{34}\,.$$

This state can be rewritten as

$$|\Phi^+\rangle_{12}\,|\Phi^+\rangle_{34} = \frac{1}{2}\Big[|\Phi^+\rangle_{14}\,|\Phi^+\rangle_{23} + |\Phi^-\rangle_{14}\,|\Phi^-\rangle_{23} + |\Psi^+\rangle_{14}\,|\Psi^+\rangle_{23} + |\Psi^-\rangle_{14}\,|\Psi^-\rangle_{23}\Big]\,.$$

This equation is the mathematical heart of entanglement swapping. If qubits 2 and 3 are measured in the Bell basis, then the result tells us which Bell state qubits 1 and 4 have become.

BSM outcome on qubits $2, 3$	Resulting state of qubits $1, 4$
$\|\Phi^+\rangle_{23}$	$\|\Phi^+\rangle_{14}$
$\|\Phi^-\rangle_{23}$	$\|\Phi^-\rangle_{14}$
$\|\Psi^+\rangle_{23}$	$\|\Psi^+\rangle_{14}$
$\|\Psi^-\rangle_{23}$	$\|\Psi^-\rangle_{14}$

Thus, the BSM outcome acts as a classical label. The quantum projection can establish entanglement between the outer qubits, but the users holding those outer qubits still need the classical BSM result to know which Bell state they share. This is why entanglement swapping does not violate special relativity: usable information still requires classical communication.

10. Using the Simulator

The simulator is organized around two main notebook cells and an optional graphics cell.

Cell 1: Tools. This cell defines the states, gates, Bell states, CNOT, Hadamard, and BSM unitary.

$$U_{\text{BSM}} = (H \otimes I)\text{CNOT}.$$

Cell 2: User interface and live graphics. The user selects one of the four Bell states and runs the BSM. The notebook displays the selected state, the state after CNOT, the state after $H \otimes I$, the measurement probabilities, the measured bitstring, the identified Bell state, and two compact live plots.

Cell 3: Optional graphics-only cell. This cell can be used to generate standalone graphics, but the recommended version places the graphics directly below the simulator output in Cell 2 so that the plots update when the input Bell state changes.

Expected behavior:

Selected Bell state	Deterministic measured bitstring
$\|\Phi^+\rangle$	00
$\|\Phi^-\rangle$	10
$\|\Psi^+\rangle$	01
$\|\Psi^-\rangle$	11

11. Guided Exercises

Exercise 1: Normalize a Bell state.

Show that

$$|\Phi^+\rangle = \frac{1}{\sqrt{2}}(|00\rangle + |11\rangle)$$

is normalized.

Solution outline.

$$\langle \Phi^+|\Phi^+\rangle = \frac{1}{2}(\langle 00| + \langle 11|)(|00\rangle + |11\rangle).$$

Using orthonormality,

$$\langle 00|00\rangle = 1, \qquad \langle 11|11\rangle = 1, \qquad \langle 00|11\rangle = 0, \qquad \langle 11|00\rangle = 0.$$

Therefore,

$$\langle \Phi^+|\Phi^+\rangle = \frac{1}{2}(1+1) = 1.$$

> **Exercise 2: Why is $|\Phi^+\rangle$ entangled?**
>
> Show why
>
> $$|\Phi^+\rangle = \frac{1}{\sqrt{2}}(|00\rangle + |11\rangle)$$
>
> cannot be written as a product state
>
> $$(a\,|0\rangle + b\,|1\rangle) \otimes (c\,|0\rangle + d\,|1\rangle).$$

Solution outline. A general product state is

$$(a\,|0\rangle + b\,|1\rangle) \otimes (c\,|0\rangle + d\,|1\rangle) = ac\,|00\rangle + ad\,|01\rangle + bc\,|10\rangle + bd\,|11\rangle\,.$$

To match $|\Phi^+\rangle$, we need

$$ac = \frac{1}{\sqrt{2}}, \qquad bd = \frac{1}{\sqrt{2}}, \qquad ad = 0, \qquad bc = 0.$$

But $ac \neq 0$ implies $a \neq 0$ and $c \neq 0$. Also $bd \neq 0$ implies $b \neq 0$ and $d \neq 0$. Then $ad \neq 0$ and $bc \neq 0$, which contradicts the required conditions. Therefore $|\Phi^+\rangle$ is not a product state and is entangled.

> **Exercise 3: Apply CNOT to $|\Phi^-\rangle$.**
>
> Starting from
>
> $$|\Phi^-\rangle = \frac{1}{\sqrt{2}}(|00\rangle - |11\rangle),$$
>
> apply CNOT and simplify.

Solution outline.

$$\mathrm{CNOT}\,|\Phi^-\rangle = \frac{1}{\sqrt{2}}(|00\rangle - |10\rangle).$$

Factor:

$$\frac{1}{\sqrt{2}}(|00\rangle - |10\rangle) = \frac{1}{\sqrt{2}}(|0\rangle - |1\rangle) \otimes |0\rangle = |-\rangle \otimes |0\rangle\,.$$

> **Exercise 4: Complete the BSM for $|\Phi^-\rangle$.**
>
> Use the result from Exercise 3 and apply $H \otimes I$.

Solution outline.

$$(H \otimes I)(|-\rangle \otimes |0\rangle) = H|-\rangle \otimes I|0\rangle .$$

Since

$$H|-\rangle = |1\rangle ,$$

we get

$$(H \otimes I)\mathrm{CNOT}\,|\Phi^-\rangle = |10\rangle .$$

Thus the measurement outcome is

$$10.$$

Exercise 5: Interpret the simulator output.

Run the simulator for all four Bell states. Record the measured bitstring for each case.

Input Bell state	Measured bitstring
$\|\Phi^+\rangle$	____________
$\|\Phi^-\rangle$	____________
$\|\Psi^+\rangle$	____________
$\|\Psi^-\rangle$	____________

Expected result.

$\|\Phi^+\rangle$	00
$\|\Phi^-\rangle$	10
$\|\Psi^+\rangle$	01
$\|\Psi^-\rangle$	11

Exercise 6: Connect to entanglement swapping.

Explain in one or two sentences why the BSM outcome must be communicated classically in an entanglement-swapping protocol.

Solution outline. The BSM outcome tells the outer parties which Bell state they share. Without that classical information, the outer parties cannot know which Bell branch was selected and cannot use or correct the entangled state in a controlled way.

12. Pre-Activity Questions

1. What are the four two-qubit computational-basis states?
2. Write the Bell state $|\Phi^+\rangle$.
3. What does it mean for a two-qubit state to be entangled?
4. What does the CNOT gate do when the first qubit is the control?

5. Why is a Bell-state measurement different from measuring each qubit directly in the computational basis?
6. In quantum networking, why might it be useful to measure two middle qubits in a joint Bell basis?

13. Post-Activity Questions

1. What unitary does the simulator apply before measuring both qubits?
2. What computational-basis state results from applying the BSM unitary to $|\Phi^+\rangle$?
3. What measured bitstring identifies $|\Psi^-\rangle$?
4. Why can the simulator identify the selected Bell state with 100% certainty?
5. Why do $|\Phi^+\rangle$ and $|\Phi^-\rangle$ require a Bell-state measurement to distinguish them reliably?
6. In entanglement swapping, what does the BSM outcome tell us about the outer qubits?
7. Why does entanglement swapping not violate special relativity?

14. Assessment

Conceptual Assessment

1. Explain what it means for information to be stored in the joint correlations of an entangled state.
2. Explain the statement: "A Bell-state measurement is Bell-state preparation run backward."
3. A student says: "The BSM creates information faster than light because it entangles distant qubits." Explain what is wrong with this statement.
4. In the simulator, the input $|\Psi^+\rangle$ always gives the output bitstring 01. What does this reveal about the relationship between the Bell basis and the computational basis after applying U_{BSM}?
5. Why is entanglement a resource for quantum networking?

Mathematical Assessment

1. Starting from
$$|\Psi^+\rangle = \frac{1}{\sqrt{2}}(|01\rangle + |10\rangle),$$
apply CNOT and then $H \otimes I$. Show that the result is $|01\rangle$.
2. Starting from
$$|\Psi^-\rangle = \frac{1}{\sqrt{2}}(|01\rangle - |10\rangle),$$
apply CNOT and then $H \otimes I$. Show that the result is $|11\rangle$.

3. Write the CNOT operator in tensor-product form and explain the meaning of each term:

$$\text{CNOT} = |0\rangle\langle 0| \otimes I + |1\rangle\langle 1| \otimes X.$$

4. Show explicitly why $|\Phi^+\rangle$ cannot be written as a product state.

15. Answer Key

Pre-Activity Questions

1. The four computational-basis states are

$$|00\rangle, \quad |01\rangle, \quad |10\rangle, \quad |11\rangle.$$

2. $$|\Phi^+\rangle = \frac{1}{\sqrt{2}}(|00\rangle + |11\rangle).$$

3. A two-qubit state is entangled if it cannot be written as a product of two independent single-qubit states. The state contains information in the joint correlations between the qubits.
4. CNOT flips the second qubit if the first qubit is $|1\rangle$, and does nothing if the first qubit is $|0\rangle$.
5. Measuring each qubit directly in the computational basis does not identify Bell phases and correlations. A BSM first changes basis from the Bell basis to the computational basis, allowing the Bell state to be identified.
6. In entanglement swapping, measuring the middle qubits in the Bell basis projects the outer qubits into a corresponding entangled state.

Post-Activity Questions

1. The simulator applies

$$U_{\text{BSM}} = (H \otimes I)\text{CNOT}.$$

2. $$|\Phi^+\rangle \to |00\rangle.$$

3. $$|\Psi^-\rangle \to 11.$$

4. The Bell states are orthonormal, and U_{BSM} maps each one to a distinct computational-basis state. Measuring in the computational basis therefore distinguishes them perfectly.
5. Direct computational-basis measurement gives the same possible outcomes for both states, 00 or 11. The difference is the relative phase between the two terms. The Bell-state measurement changes basis so that this phase difference becomes a different computational-basis bitstring.
6. The BSM outcome tells us which Bell state the outer qubits share.

7. Entanglement swapping does not transmit usable information faster than light because the BSM result must be communicated through an ordinary classical channel.

Conceptual Assessment

1. In an entangled state, neither qubit alone carries the full information. Instead, the important information appears in the correlations between measurement outcomes of the two-qubit pair.
2. Bell-state preparation maps computational-basis states into Bell states. A Bell-state measurement applies the inverse map, converting Bell states back into computational-basis states that can be read out by ordinary measurement.
3. The quantum state description changes conditionally, but usable information about the outcome requires a classical message. Since that classical message cannot travel faster than light, relativity is not violated.
4. It shows that the inverse Bell circuit maps the Bell basis onto the computational basis one-to-one. The Bell label becomes a two-bit classical measurement result.
5. Entanglement is a resource because it creates nonclassical correlations between separated systems. These correlations can support teleportation, entanglement swapping, quantum repeaters, distributed sensing, and networked quantum information processing.

Mathematical Assessment

1. Start with
$$|\Psi^+\rangle = \frac{1}{\sqrt{2}}(|01\rangle + |10\rangle).$$
Apply CNOT:
$$\mathrm{CNOT}\,|01\rangle = |01\rangle\,, \qquad \mathrm{CNOT}\,|10\rangle = |11\rangle\,.$$
Therefore,
$$\mathrm{CNOT}\,|\Psi^+\rangle = \frac{1}{\sqrt{2}}(|01\rangle + |11\rangle).$$
Factor:
$$\frac{1}{\sqrt{2}}(|01\rangle + |11\rangle) = \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle) \otimes |1\rangle = |+\rangle \otimes |1\rangle\,.$$
Apply $H \otimes I$:
$$(H \otimes I)(|+\rangle \otimes |1\rangle) = |0\rangle \otimes |1\rangle = |01\rangle\,.$$
2. Start with
$$|\Psi^-\rangle = \frac{1}{\sqrt{2}}(|01\rangle - |10\rangle).$$
Apply CNOT:
$$\mathrm{CNOT}\,|\Psi^-\rangle = \frac{1}{\sqrt{2}}(|01\rangle - |11\rangle).$$
Factor:
$$\frac{1}{\sqrt{2}}(|01\rangle - |11\rangle) = \frac{1}{\sqrt{2}}(|0\rangle - |1\rangle) \otimes |1\rangle = |-\rangle \otimes |1\rangle\,.$$

Apply $H \otimes I$:

$$(H \otimes I)(|-\rangle \otimes |1\rangle) = |1\rangle \otimes |1\rangle = |11\rangle\,.$$

3. The operator

$$\text{CNOT} = |0\rangle\langle 0| \otimes I + |1\rangle\langle 1| \otimes X$$

means: if the first qubit is $|0\rangle$, apply I to the second qubit; if the first qubit is $|1\rangle$, apply X to the second qubit.

4. A general product state is

$$(a|0\rangle + b|1\rangle) \otimes (c|0\rangle + d|1\rangle) = ac|00\rangle + ad|01\rangle + bc|10\rangle + bd|11\rangle\,.$$

To equal

$$|\Phi^+\rangle = \frac{1}{\sqrt{2}}(|00\rangle + |11\rangle),$$

the coefficients of $|01\rangle$ and $|10\rangle$ must vanish:

$$ad = 0, \qquad bc = 0,$$

while the coefficients of $|00\rangle$ and $|11\rangle$ must be nonzero:

$$ac = \frac{1}{\sqrt{2}}, \qquad bd = \frac{1}{\sqrt{2}}.$$

These requirements cannot all be satisfied simultaneously. Therefore $|\Phi^+\rangle$ cannot be written as a product state and is entangled.

16. Takeaway

Bell states are maximally entangled two-qubit states. Their information is stored not in either qubit alone, but in the joint quantum correlations between the two qubits. This makes them essential resources for quantum communication and quantum networking.

The Bell-state measurement simulator demonstrates how these entangled states can be distinguished. By applying the inverse Bell-preparation circuit,

$$U_{\text{BSM}} = (H \otimes I)\text{CNOT},$$

the simulator maps the Bell basis onto the computational basis:

$$|\Phi^+\rangle \to |00\rangle\,, \qquad |\Phi^-\rangle \to |10\rangle\,, \qquad |\Psi^+\rangle \to |01\rangle\,, \qquad |\Psi^-\rangle \to |11\rangle\,.$$

A final computational-basis measurement then produces two classical bits that identify the Bell state.

A Bell-state measurement converts entangled Bell-basis information into a classical two-bit label.

For a standalone Bell state, that label identifies the input state. For entanglement swapping, the same measurement logic labels the entangled state created between two outer qubits. That classical label is essential: without it, the outer parties do not know which Bell state they share.

This is why the BSM simulator is a natural prelude to an entanglement-swapping simulator. It teaches the central measurement mechanism first, before adding the larger network protocol.